Deterministic Interpolation

HES 505 Fall 2025: Session 19

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Objectives

- **Describe** reasons for interpolating data
- **Distinguish** between deterministic and probabilistic methods for interpolation
- **Characterize** the difference between distance- and spline-based interpolation
- **Implement** two common interpolation methods in R

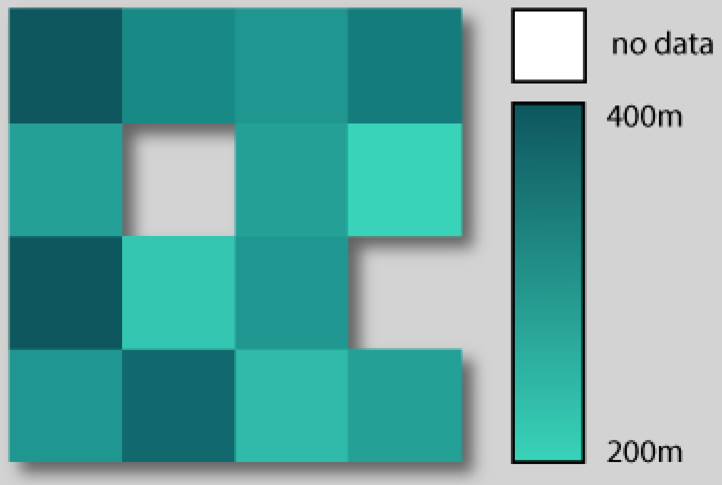
How do you view the world?

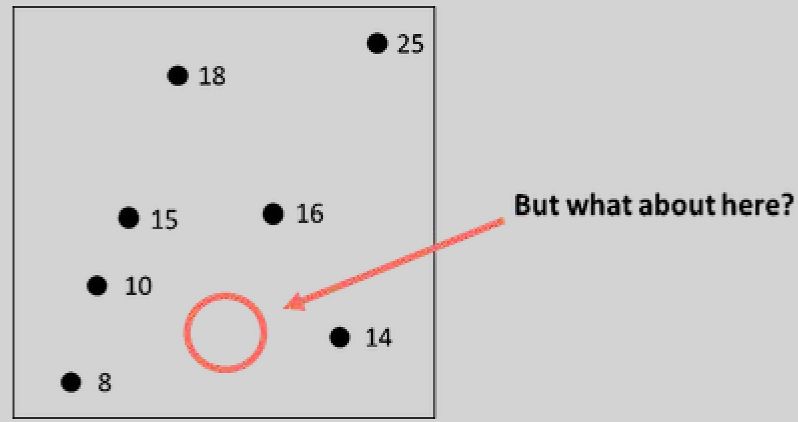
…As a Continuous Field

::: columns ::: {.column width="50%"} :::{style="font-size: 0.8em;"} - The earth is a single entity with properties that vary continuously through space

- **Spatial continuity: Every cell has a value (including “no data” or “not here”)**
- Self-definition: the values define the field{.semi-transparent}
- Space is tessellated: cells are mutually exclusive :::

…but what about missing data?


no data
400m
200m


25
18
15
16
10
14
8
But what about here?

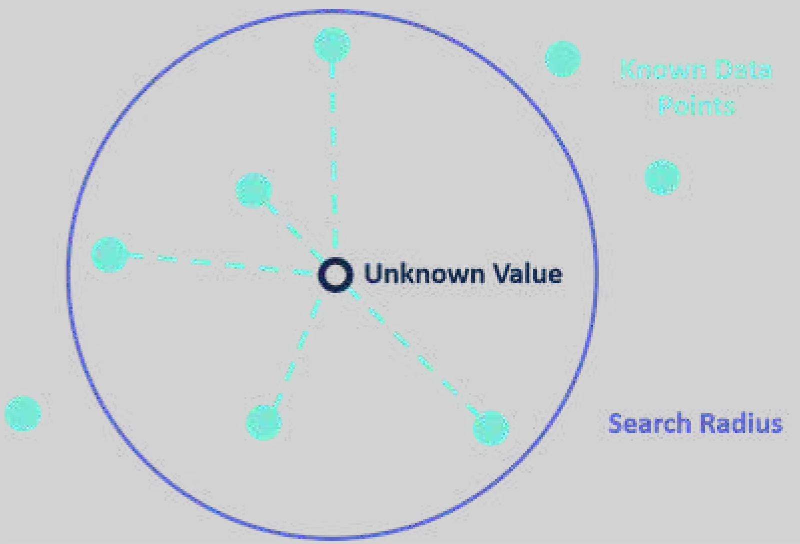
Spatial data as a stochastic process

$$Z(\mathbf{s}) : \mathbf{s} \in D \subset \mathbb{R}^d$$

Interpolation

- Goal: estimate the value of Z at new points in **D**
- Deterministic: Surfaces created directly from measured points
- Probabilistic: Relies on statistical properties of measured points

Inverse-Distance Weighting



- Unknown point estimated as weighted average
- Weight based on distance (Tobler)
- Can include all points or a subset

Inverse-Distance Weighting

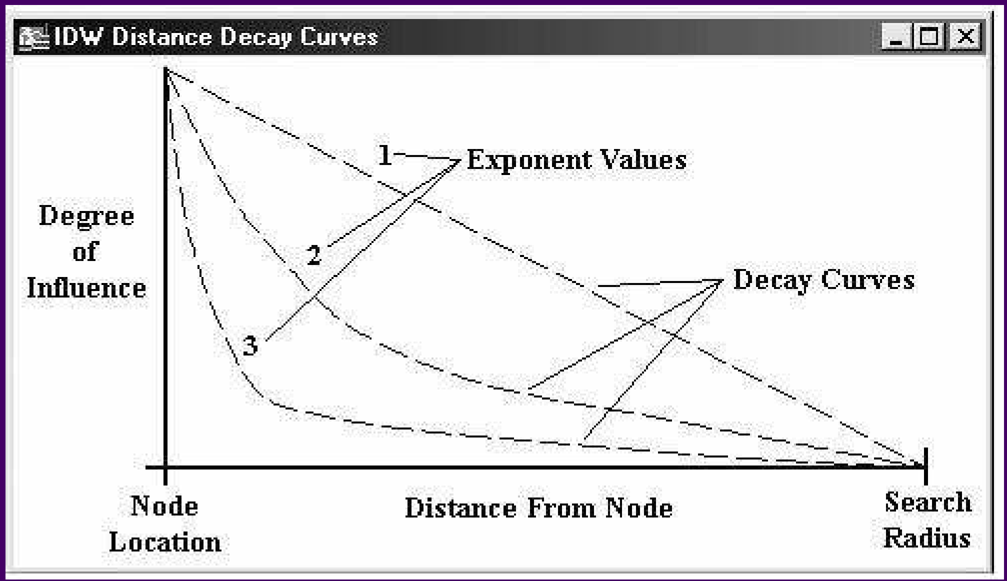
$$\hat{z}(\mathbf{x}) = \frac{\sum_{i=1} w_i z_i}{\sum_{i=1} w_i}$$

where

$$w_i = |\mathbf{x} - \mathbf{x}_i|^{-\alpha}$$

α is known as the *power* and controls the weight.

Thinking about Power

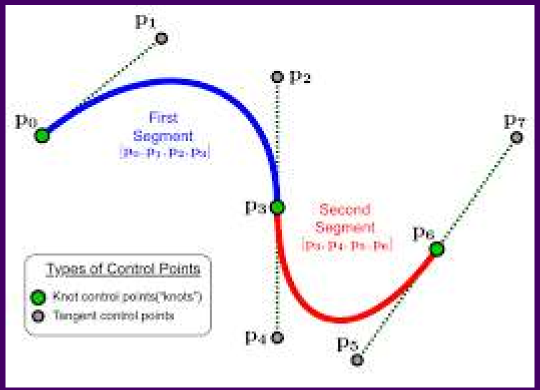


- Decay = negative exponent (1=linear, 2=quadratic, etc)
- Larger values = more localized results
- Lower values = more smoothed (averaged) results

IDW and you

- Assumes linear relationship between distance and the unknown value
- Bounded estimation - can't predict values it's never seen
- Interpolated values are local-ish averages
- Particularly useful for dense, equally distributed data

Splines

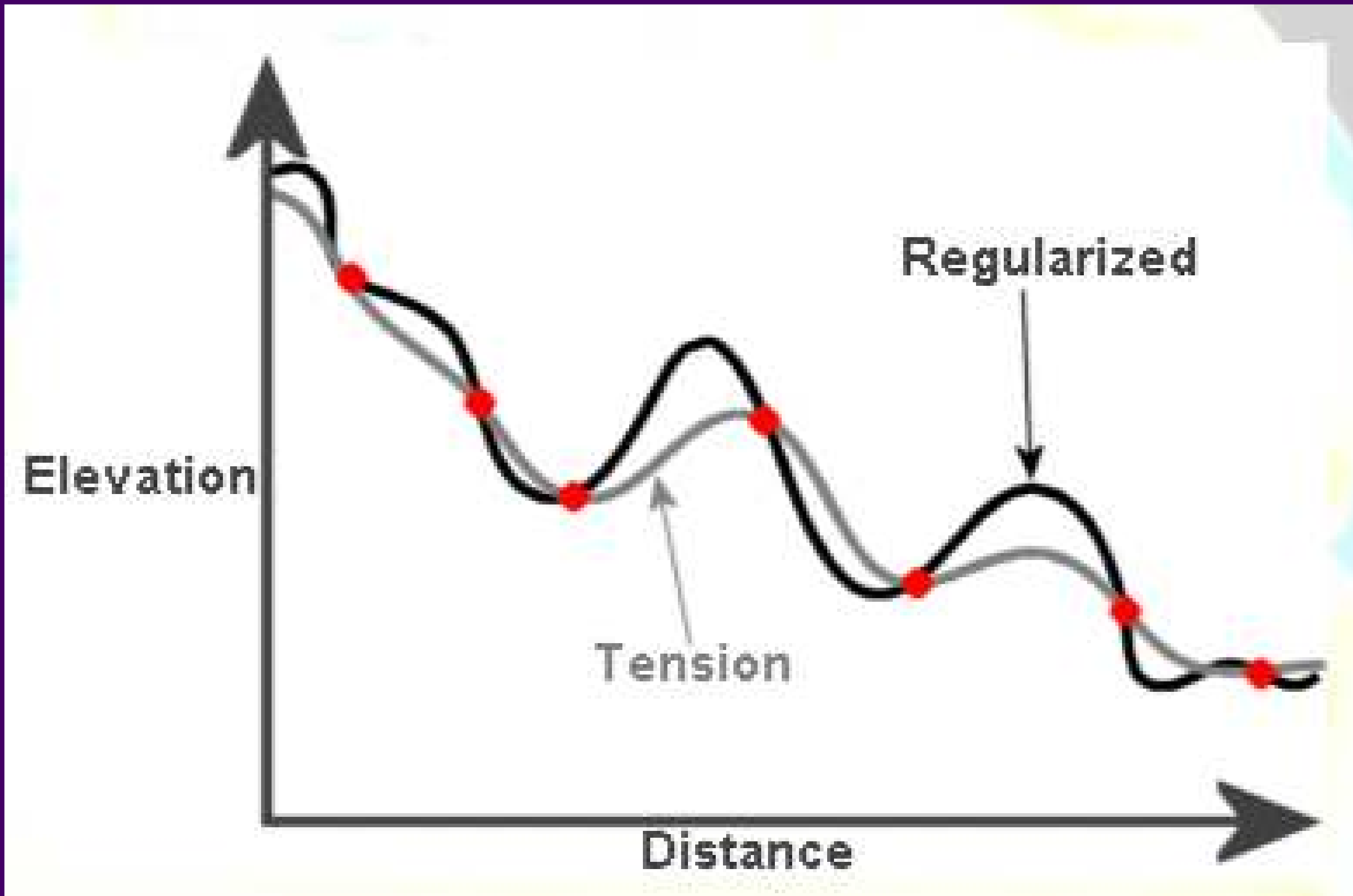


- A mathematical function for a curve that passes near / through points
- Piecewise polynomials that are smoothed between endpoints
- Allows for nonlinearity in relationships between distance and unknown values

Splines, so many

- Tension Spline: flatter, forces estimated values to stay closer to control points
- Regularized Spline: more elastic, more likely to produce values above and below the control points
- Thin-plate Spline: a bit of middle-ground between the two.

Splines, so many



Splines and You

- Great when you know you've undersampled the extreme data points
- Smooth, continuous surface is the goal
- Bad when close points have very different values